Module 1 Lecture - Basics of Sampling and Data

Introduction to Statistical Methods

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1 Overview and Introduction

1.1 Textbook Learning Objectives

- Recognize and differentiate between key terms used often in statistics, probability, and data
- Apply and identify various types of sampling methods, and how they impact the broader meaning of the data
- Create and interpret frequency/contingency tables.

1.2 Instructor Learning Objectives

- Understand how variables can be described and represented with different scales
- Appreciate the basic of how data is collected, and how the collection method implies the scale of measurement

1.3 Introduction

- Statistics comprises various quantitative methods to ______ and analyze numeric and categorical data
  - First and foremost, it is technically a subfield of ______
  - I like to think of statistics as a set of ______ methods to research various sciences - biology, psychology, education, economics, etc.
- At the core of statistics is probability, or how we describe the ______ of certain events occurring
  - We'll briefly dive into the ______ of probability in module 3 and later expand on that with ______ statistics
- In the modern age, statistics often tends to be closely associated with fields like ______ science and computer science
  - It's important to understand that how we gather and transform data has a ______ impact on the end result on our analyses
  - Our analyses are both ______ and ______ by what we do to gather and transform our data
- Statistics are just one part of the total research process, but are affected greatly by how we ______ our studies
  - In fact, statisticians are often involved with the entire research process, not just ______ - because they need to advise on the "best" way to gather data so that it is useful

> **Important**
>
> While this is not a research design class, we will talk about design as it is relevant to analysis; problems in design can be very problematic for the meaning of the statistics!

- As we progress through this class, we'll also touch on how to ______ and show-off data and results
  - Especially as we use more complicated methods, using the right graphical techniques can help us make our results ______ to non-statistical audiences.
- To start we need to establish many of the words we often use in statistics, across the topics we'll be covering in this semester

> **Important**
>
> Many terms in statistics have alternative names (which can be frustrating on occasion), I'll try to highlight when a term may sometimes be called something else.

2 Definitions of Statistics, Probability, and Key Terms

2.1 Introduction

- Statistics starts with data, which is some ______ amount of a phenomenon
  - In ______ research, our data may be grades, ______ of student well being, measures of student engagement, etc.
  - At a more macro-scale, universities use measures like enrollment, class size, financial cost, etc for analysis
  - The word "measure" is important here - we'll come back to that

> Discuss: What are other examples of common data we would gather in education?

- We can treat this data via $\boxed{\text{descriptive statistics}}$, where we summarize, describe, and demonstrate various components and structure of the data
  - Examples of use:
    * What is the most commonly occurring letter grade in a course?
    * What is the average student GPA in each major?
    * Is the family income of students at this university skewed lower or higher?
  - Types of descriptive statistics: ____________, mode, skew, standard deviation, etc. - we'll describe all of these in more detail later
- Usually in addition to descriptives, we would also do some type of $\boxed{\text{inferential statistics}}$, which helps draw probabilistic conclusions from the data
  - Examples of use:
    * Are students in these two classrooms significantly different in their standardized test scores?
    * Has there been a significant improvement in self-reported teacher well-being in the last 3 years?
  - E.g., T-tests, ANOVA, anything that gives us ____________ - once again, we'll be revisiting what exactly these are later in the course

> Discuss: Take the example of this question: What are the counts of parents who approve or disapprove of their gifted child's treatment in school? Does this seem like a question to address with descriptive or inferential statistics?

- Crucially, statistics should always be ____________ by a desire theoretically understand and investigate data
  - As researchers, our goal is to not ____________ for conclusions from data, it is to carefully and responsibly treat it for analysis
  - Metaphorical example: we are not casting a wide net, we are casting a single line towards an area we are interested in

2.2 Probability

- Probability helps us contextualize inferential statistical ____________ as having some amount of ____________ that our results occurred not simply due to chance

- Really, we want good, statistical evidence that our results represent something *real*

> Important
>
> This is why we spend nearly the first half of this semester just establishing the role of probability and distributions before introducing inferential tests!

- Probability underlies many __________ we make intuitively
  - E.g., What are the chances the car I buy will be low maintenance costs? How likely am I to really use this new device I buy? If I get this degree, I think I should have a higher likelihood of getting my desired job

> Discuss: Give another example of considering probability of a certain event occurring in your personal life

- Just like with our personal life, probability can be a useful way to understand a certain __________ in our data and research, and measure our confidence
  - E.g., What are the chances that if I took a new sample of students, I'd see this same average GPA? If I compare these two schools in resulting annual income of graduates, what are the chances my results would be the same if replicated?

3 Sampling

3.1 Start with Sampling

- When we do research, the group we desire to __________ and understand is our population of interest
  - E.g., *All* high school students, *all* teachers
  - The population is usually a group we can't practically, __________ study (as they are usually too large or dispersed!)
  - In the case that we did somehow gather data about __________ members of a population, we would call this a census

Discuss: Given this definition of a census, I just provided, explain why would we call the periodic counting of all individuals in the US a 'census'

- Instead of gather data on the population as a whole, we take a __________ subset of individuals that are meant to represent the population, which we would call a sample
  - We get our sample via some method of sampling, which is exactly how we get our subset - it can be done in a "good" or "bad" way, more on this later
- The numeric values that we use to calculate and describe on our __________ is called a statistic, which, in turn, is meant to represent the parameter of the population
  - Another way to say this is that the sample statistic is an __________ of the population parameter
  - A mnemonic to remember this:
    - **P**oulation $\longrightarrow$ Parameter
    - **S**ample $\longrightarrow$ Statistic
- But, remember our goal is often to study the __________, not just the subset we gather data for!
  - Thus, we want to have great __________ that whatever statistics we have, __________ represent their respective parameters
  - One part of ensuring this accuracy is to use a sampling method that results in a representative sample

Important

Don't lose sight of this - our sample is just a practical way for us to try and say something about our population, which is what we are really after.

3.2 Better Sampling

Important

This section is going to briefly veer into good research design, as it plays an important role in what conclusions we can make from the data. Our statistics are only as valuable as how well they represent their parameters!

- Ideally, a __________ sample will have similar characteristics in similar proportions to the population they are taken from
  - E.g., veterans in the United States skew more male than female, thus, a representative sample of that population will likely be more men than women
  - A "good" sampling method, should result in this __________ between the sample and population
- Good sampling methods are those that are random, which means that each member of the population has an equal, __________ chance of being selected into the sample
  - E.g., if my population was veterans, and there are 1000 total veterans, a random sampling technique would mean that each of those 1000 individuals has a 1 in 1000 chance of being in the sample

Discuss: I am interested in studying visitors to the local mall. I catch people entering at one entrance to the mall (there are 4 entrances total) and ask them several questions. Does this meet the definition of a random sample as stated above?

- Within the broader classification of __________ sampling methods, there are several subtypes
  - Simple random sampling is when we __________ all individuals in the population and simply use some random method to select numeric IDs of individuals to be included in the sample
  - Systemic sampling is similar, but uses a process of starting with a specified ID and __________ by a specific number to select the others

- Cluster sampling is done when we have pre-existing clusters, or ______, in our population, and we randomly select from those clusters to include all individuals from the selected clusters
- Stratified sampling is similar in concept to cluster sampling, but done with some individualistic ______ of the participants

> **Important**
>
> There are many more types of sampling that can be covered in a research methods class

- Any **non-random** method that does not follow the above philosophy can not ensure the ______ of the sample
  - Our ______ and interpretation may be swayed or hindered by this non-randomness, thus causing a sampling error
  - Another way to describe this is by calling this ______ process as a sampling bias

> Discuss: My population of interest is all education psychology students. I put up a poster with a QR code in the physical office for the educational psychology department, and wait for individuals to complete my survey, does this seem like a random or non-random method of sampling?

3.3 Natural Variation Between Samples

- It is normal and ______ that, if we take two different samples of a population, we can and will find that they have somewhat different characteristics
- However, if these are both believed to be representative samples, they should become ______ similar as they become larger
  - This is an idea we will revisit in our discussion on the Central Limit Theorem, a concept we'll go into much more later

- Proper use of statistics will help us __________ for the fact that our samples naturally vary

> **Important**
>
> One could say that the entire goal of inferential statistics is in controlling for variation between samples!

4 Descriptions of Data

4.1 Introduction

- Variables are some defined measure/observation with variance in the data
  - I.e., there needs to be __________ numbers in the data, the characteristic can take different values - otherwise it is a statistical constant
  - __________ can also be either numeric or categorical, that is, they produce data of a specified type
  - E.g., Age in years → __________
  - E.g., Job title → __________

> **?** I gather information about all the individuals enrolled at a local college, and all live in the state of Indiana. Is state of residence a constant or variable, and is it categorical or numeric?
>
> - (●) A) Categorical Constant
> - ( ) B) Categorical Variable
> - ( ) C) Numeric Constant
> - ( ) D) Numeric Variable
>
> Explanation:

- Let's dive more into the different __________ of data
  - These are closely related to the above "categorical" and "numeric" terms
- qualitative data come from a more descriptive (via words or __________)
  - E.g., Eye/hair color, more complicated: description of internal feelings
  - Because of it's nature, it is almost always __________ in nature

  - Qualitative data is often times represented in ____________ of occurrences of a certain description, for the purpose of analysis - e.g., I have 10 people in my sample with brown eyes, and 5 with blue eyes
- quantitative data is something represent by an ____________ or numeric measurement
  - However, within quantitative data, it can be discrete or continuous
  - Discrete data is that which is ____________, or has no intervals between the integers, e.g., number of phone calls had → I can't have half a phone call
  - Continuous data does indeed have ____________ between integers, e.g., Age → I can be half a year of age.
  - For better or for worse, these two types are often treated ____________, even when that may not be accurate

> **?** Consider this variable: Numbers of children a parent has. What could this be classified as?
>
> - (●) A) Qualitative
> - ( ) B) Discrete Quantitative
> - ( ) C) Continuous Quantitative
> - ( ) D) None of the above
>
> Explanation:

> **!** Important
>
> As a general rule of thumb, quantitative data is easier and more versatile in analysis, but also is more reductive. This class will focus more on quantitative data, but I encourage you to look at other courses that are more qualitative in nature!

4.2 More on Qualitative Data

- As alluded to early, qualitative data can be represented by counts/proportions
  - These are often shown in tables or graphs that somehow ____________ those proportions
  - When dealing especially with categorical data, it can be useful to repre-

sent the data in terms of how often a certain data value ______ and how often it occurs relative to the other possibilities.

Gender	Frequency	Relative Frequency
Men	18	0.45
Women	22	0.55
Total	**40**	**1.00**

- Frequency is the raw number of times that a certain characteristic or value occurs, whereas Relative frequency

5 Levels/Scales of Measurement

5.1 Introduction

- Earlier, we gave some examples of how to ______ different variables as qualitative, quantitative, discrete, continuous, etc.
- We'll introduce some other terms to help us classify our variables, which will be ______ when we treat it

> **Important**
>
> It cannot be understated how important it is that you can properly describe and classify your variables, as it is crucial in determining what descriptive and inferential statistics can be used.

5.2 Levels of Measurement

- Nominal scale variables are qualitative and categorical, with classifying ______ and no defined "order".
  - E.g., state/country of residence, hair/eye/skin color
- Ordinal scale variables are those that have an order, but there is not a clear ______ between each interval or place in the order
  - Thus, it sort of blurs the line between categorical and ______
  - E.g., place in a foot race, class rank
- Interval scale variables do have a clear, consistent interval in-between each integer, but no absolute zero point

- E.g., temperature, IQ scores, ranges, etc.

- Ratio scale is the same as interval scale, but does have a clear zero point as well.
  - E.g., score on an exam
- Generally, the different levels/scales of measurement are different levels of restrictive for analysis, in this order: Nominal (most restrictive) to Ratio (least restrictive).
  - Of course, having mostly nominal scale data does not always spell doom, but it can involve more tricky ________________

Discuss: I put all the students in my class from shortest to tallest and assign them the number they are from being the shortest in the class, what scale of measurement would this data be and why?

6 Conclusion

6.1 Recap

- Using statistics with our data first starts with properly understanding and identifying our variables, describing them and their scales of measurements
- Sampling plays an important role in how representative our data is of the population of interest, which, in turn, helps establish whether our results are generalizable or not
- Probability is at the core of inferential analyses, and helps us establish the likelihood of events occurring, which we will be revisiting later

Key Terms

C

categorical A description of data in that it can be represented with qualitative labels and represents some type of group membership 9

census A scenario in which we gather data on all members of a population of interest 5
Cluster sampling Sampling, at random, from pre-existing groups of people 8
continuous A description of numeric data that implies intervals in-between each integer 10
counts/proportions A way to describe data that transforms qualitative/categorical data into a number of occurrences 10

D

descriptive statistics A type of statistics employed for the purpose of summarizing and describing variables/data 4
discrete A description of numeric data that implies ’counts’, i.e., data that can not have parts in-between 10

F

Frequency The raw number of times that a certain characteristic or value occurs 11

I

inferential statistics A type of statistics employed to determine the rarity and probability of certain data results occurring 4
Interval scale Scale of measurement that has clear distance between each point, but no clear zero point 11

N

Nominal scale Scale of measurement that has no order and no distance between categories 11
numeric A description of data in that it can be represented with numbers 9

O

Ordinal scale Scale of measurement that has inconsistent or unclear distance between each point, but does have order 11

P

parameter A number or value in the population, estimated via the sample statistic 6
population of interest A group of individual with specified characteristics, that we are interested in studying and producing results that are generalizable to the broader group 5
probability The study of likelihood of events occurring under certain circumstances 2

Q

qualitative data Data/variables that describe a characteristic in words 9

quantitative data Data/variables that describe a characteristic in numbers 10

R

random A probability event in which each outcome has an equal chance of occurring 7
Ratio scale Scale of measurement that has clear distance between each point AND a clear zero point 12
Relative frequency The relative percentage or ratio of times a certain value occurs relative to the other possibilities 11
representative sample A sample which is genuinely representative of the population of interest it is pulled from; usually taken by some form of random sampling 6

S

sample A subset of individual taken from the population of interest, but are meant to be representative of the broader population. Those who data is actually gathered from. 6
sampling The process by which we select individuals from the population to be included in our sample 6
sampling bias A scenario in which the method by which sampling was done has resulted in a non-representative sample 8
Simple random sampling All individuals have unique IDs are chosen completely at random 7
statistic A calculated number or value taken from our sample, meant as an estimate of the population parameter 6
statistical constant Some characteristic or measurement that only has one constant value; opposite of variable 9
Statistics A subfield of math that focuses on the application on quantitative methods to describe and analyze numeric and categorical data, through the lens of probability 2
Stratified sampling Similar to cluster sampling, but with a characteristic of a person 8
Systemic sampling Staring a certain ID of person and iterating by a specified number 7

V

Variables Measurements or characteristics that change or can take different values 9

The instructor-provided glossary may not include all terms worth memorizing, make sure you consider using the vocabulary list in you book and your own judgment to make sure you have all relevant terms